HDOC Day-73

Implementation of Enumerations for Days and Traffic Lights in C

Aim

To implement two C programs using enumerations: print the seven days of the week with their default integer values, and display Stop, Wait, or Go for a given traffic-light signal.

Algorithm - Program 1: Days of the Week

1. Start the program and define enum Days with SUNDAY through SATURDAY.
2. Allow the default integer values to begin at 0 and increase consecutively.
3. Print each named day with its corresponding integer value.
4. Stop the program.

Algorithm - Program 2: Traffic Lights

1. Define enum TrafficLight containing RED, YELLOW, and GREEN.
2. Read the traffic-light name as a string.
3. Compare the input with the three valid names and assign the enum variable.
4. Use switch to display Stop for RED, Wait for YELLOW, or Go for GREEN.
5. Print an error for invalid input and terminate.

C Program 1 - Enumeration for Days

```
#include <stdio.h>

enum Days {
    SUNDAY, MONDAY, TUESDAY, WEDNESDAY,
    THURSDAY, FRIDAY, SATURDAY
};

int main()
{
    printf("Days of the Week and Their Integer Values:\n\n");
    printf("SUNDAY = %d\n", SUNDAY);
    printf("MONDAY = %d\n", MONDAY);
    printf("TUESDAY = %d\n", TUESDAY);
    printf("WEDNESDAY = %d\n", WEDNESDAY);
    printf("THURSDAY = %d\n", THURSDAY);
    printf("FRIDAY = %d\n", FRIDAY);
    printf("SATURDAY = %d\n", SATURDAY);
    return 0;
}
```

C Program 2 - Enumeration for Traffic Lights

```
#include <stdio.h>
#include <string.h>

enum TrafficLight { RED, YELLOW, GREEN };

int main()
{
    enum TrafficLight signal;
    char input[20];

    printf("Enter traffic light (RED/YELLOW/GREEN): ");
    scanf("%19s", input);

    if (strcmp(input, "RED") == 0)
        signal = RED;
    else if (strcmp(input, "YELLOW") == 0)
        signal = YELLOW;
    else if (strcmp(input, "GREEN") == 0)
        signal = GREEN;
    else {
        printf("Invalid traffic light signal!\n");
        return 0;
    }

    switch (signal) {
        case RED:    printf("Stop\n"); break;
        case YELLOW: printf("Wait\n"); break;
        case GREEN:  printf("Go\n"); break;
        default:     printf("Invalid signal!\n");
    }
    return 0;
}
```

Explanation

An enumeration (enum) is a user-defined type whose named constants have integer values. Unless explicitly assigned, the first constant has value 0 and subsequent constants increase by 1. The Days enumeration therefore maps SUNDAY to 0 through SATURDAY to 6. The TrafficLight enumeration maps RED to 0, YELLOW to 1, and GREEN to 2. strcmp() identifies the input signal, and switch selects the corresponding instruction.

Days and Integer Values

Day	Integer value
SUNDAY	0
MONDAY	1
TUESDAY	2
WEDNESDAY	3
THURSDAY	4
FRIDAY	5
SATURDAY	6

Traffic Light Mapping

Signal	Value	Instruction
RED	0	Stop
YELLOW	1	Wait
GREEN	2	Go

Test Case Table

Test	Program / Input	Expected result
1	Days / no input	All days displayed, 0 through 6
2	Traffic / RED	Stop
3	Traffic / YELLOW	Wait
4	Traffic / GREEN	Go
5	Traffic / BLUE	Invalid traffic light signal!

Sample Output - Program 1

```
Days of the Week and Their Integer Values:

SUNDAY = 0
MONDAY = 1
TUESDAY = 2
WEDNESDAY = 3
THURSDAY = 4
FRIDAY = 5
SATURDAY = 6
```

Sample Output - Program 2

```
Enter traffic light (RED/YELLOW/GREEN): GREEN
Go

Enter traffic light (RED/YELLOW/GREEN): RED
Stop

Enter traffic light (RED/YELLOW/GREEN): YELLOW
Wait

Enter traffic light (RED/YELLOW/GREEN): BLUE
Invalid traffic light signal!
```

Result

Both C programs demonstrate the use of enum constants. The first prints the seven days and their integer values from 0 to 6; the second displays the appropriate Stop, Wait, or Go instruction for the selected traffic-light signal and handles invalid input.